UNIVERSIDAD POLITÉCNICA DE MADRID
ESCUELA TÉCNICA SUPERIOR DE INGENIEROS INFORMÁTICOS



Explainable Classification of Payment Difficulty

Inteligencia Artificial Explicable: estudio de caso

Thales S. A. Camargo
Jhon Corrales Almendariz
Joel Montes de Oca Martínez

Summary of results

Five-fold CV AUC of each model



DECISION TREE

17 leaves, 6 rules
Accuracy 0.812
F1 0.49 at t_bal



LOGISTIC REG.

Log of money vars.
Accuracy 0.814
F1 0.50 at t_bal



EBM

Best in 5/5 folds
Accuracy 0.817
F1 0.52 at t_bal

Same 80/20 train/validation split and the same five CV folds for every choice: hyperparameters, a log transform and the thresholds (t_acc for accuracy, t_bal for balanced accuracy). All models beat the majority baseline (accuracy 0.788); a black-box gradient-boosting model gets AUC 0.768. Submitted: the EBM.

DECISION TREE

17 leaves that collapse into six rules

Flag difficulty if any rule fires (status = latest repayment status, ratio = payment-to-bill ratio):

R1 status ≥ 1 and ratio ≤ 0.095: risk 0.54; 15 % of clients, 39 % of all difficulties

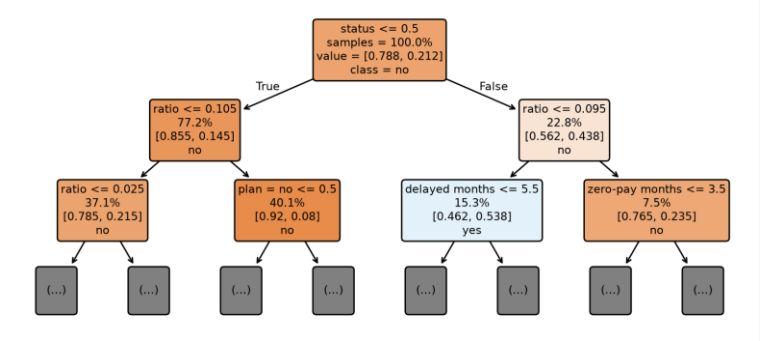
R2 status ≤ 0 and ratio ≤ 0.025: 0.43

R3 status ≥ 1, ratio > 0.095, zero-payment months ≥ 4: 0.33

R4–R6 low ratio with age ≤ 24 or with an assistance plan: 0.26–0.32

Global: uses 7 of 15 variables; status and ratio give 85 % of its importance.

Trade-off: easiest to apply by hand, but coarse (17 probability levels): CV AUC 0.752, the lowest.



Local case C (actual: difficulty)

Path: status ≥ 1 → ratio > 0.095 → zero-payment months ≤ 3 → ratio > 0.315: leaf risk 0.15, missed.

A ratio of 4.39 on a negative bill looks safe to the tree; LR (0.50) and the EBM (0.55) flag the client.

LOGISTIC REGRESSION

Compact and stable, but conditional and linear

Setup: standardized inputs; a CV-chosen log of the money variables adds 0.012 AUC.

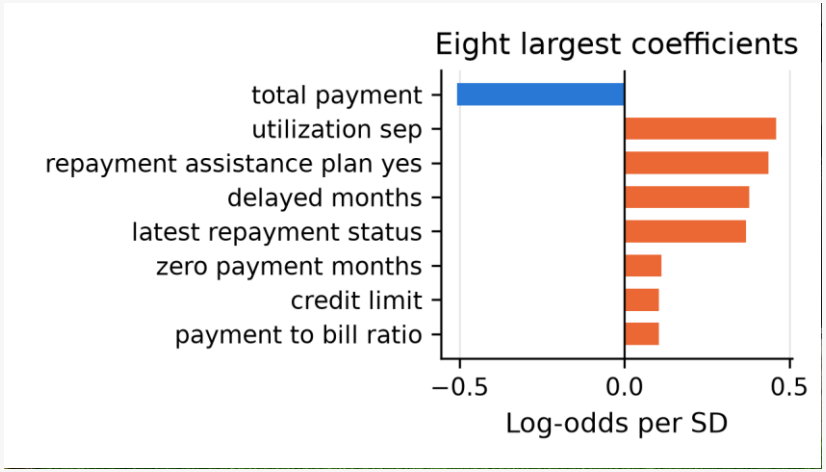
Global: risk rises with utilization, the assistance plan, delayed months and repayment status; it falls with total payment.

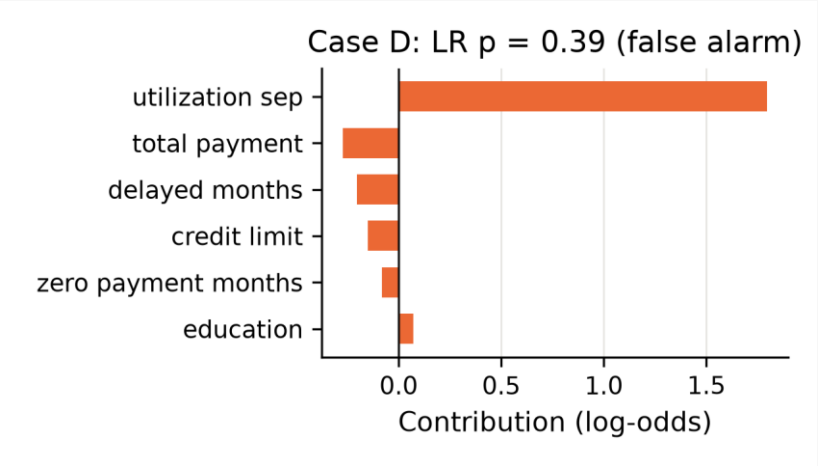
Caution: coefficients are conditional: maximum delay, credit limit and payment ratio flip their sign (collinearity).

Limit: straight lines only; status codes −2, −1 and 0 are forced onto a line.

Local, case D: utilization 2.0 adds +1.80: p = 0.39, a false alarm. The tree ignores utilization and is right.

Performance: CV AUC 0.761; F1 0.50 at t_bal.





EBM

The best model: CV AUC 0.776

Ahead of LR and the tree in all five CV folds; submitted for test.csv (t_acc = 0.49).

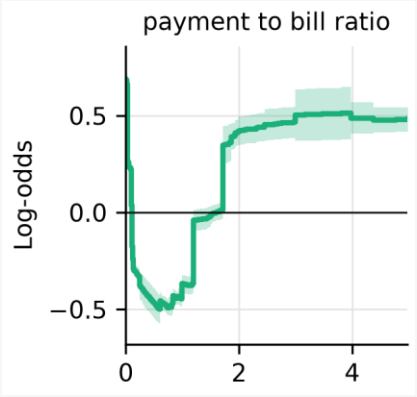
Additive: one shape function per variable (interactions = 0), read directly from the plots.

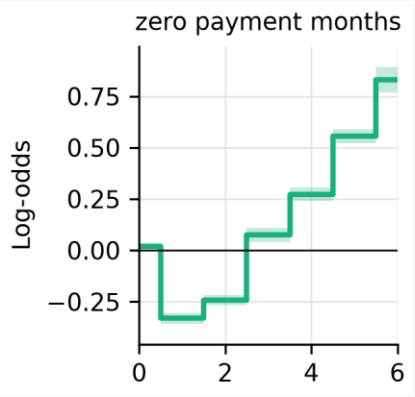
Payment ratio is U-shaped: risk is lowest at 0.5–1.2 and rises again above 1.7 (catch-up payments, tiny bills).

Zero-payment months: 1–2 lower the risk, 3+ raise it. Forcing a monotone shape lowers the CV AUC (0.7756 → 0.7728), so the dip is real.

Age is U-shaped: clients under 25 and over 57 are riskier.

Assistance plan +0.41: a marker of earlier stress, not a causal harm.





Local case C

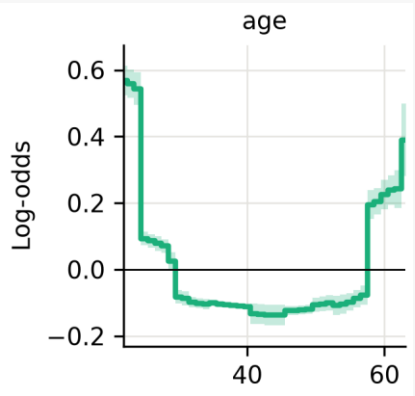
Ratio 4.39: +0.48

Delays: +0.49

Plan: +0.41

p = 0.55: flagged

(tree: 0.15, missed)




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